



中国认可
国际互认
检测
TESTING
CNAS L4136



检测报告 TEST REPORT

报告编号/ Report No..... : WTF24H04091192R1C

委托方/ Applicant : 南京永立电子有限公司
Nanjing Yongli Electronics Co., Ltd.

地址/ Address : 南京市江宁高新园天册路 6 号
No. 6, Tiance Road, Jiangning Science Park, Nanjing City

样品名称/ Sample Name : 导电高分子固态铝电解电容/ Conductive polymer solid
aluminum electrolytic capacitor

检测要求/ Test Requested : 参照欧盟 RoHS 指令 2011/65/EU 及其修订指令 EU
2015/863&客户限值, 检测样品中的铅、镉、汞、六价铬、多溴
联苯、多溴二苯醚、DBP、BBP、DEHP、DIBP 的含量。/
Determine the Pb, Cd, Hg, Cr(VI), PBBs, PBDEs, DBP, BBP,
DEHP, DIBP content in the sample with reference to EU RoHS
Directive 2011/65/EU and its amendment Directive EU
2015/863&Client's Limit.

检测结论/ Test Conclusion : 合格/ PASS

样品接收日期/ Date of Receipt sample : 2024-04-22

检测周期/ Testing period : 2024-04-22 ~ 2024-04-26

报告日期/ Date of Issue..... : 2024-04-26

检测结果/ Test Result..... : 参见报告下页/ Refer to next page (s)

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授权签字人

Signed for and on behalf of
Shenzhen Hongcai Testing Technology Co.,
Ltd.



黄胜明/Michael Huang

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HCT/RF-23-19

检测结果/ Test Result(s):

欧盟 RoHS-铅、镉、汞、六价铬、多溴联苯、多溴二苯醚、DBP、BBP、DEHP、DIBP 的含量/ EU RoHS-Pb, Cd, Hg, Cr(VI), PBBs, PBDEs, DBP, BBP, DEHP, DIBP content

检测方法/仪器/ Test Method/Equipment: IEC 62321-5:2013, IEC 62321-4:2013+AMD1:2017, IEC 62321-7-2:2017, IEC 62321-6:2015, IEC 62321-8:2017; ICP-OES, UV-VIS, GC-MS

检测项目 Test Item(s)	单位 Unit	方法检出限 MDL	结果 Result(s)		限值 Limit
			1-2	1-3	
铅 Lead(Pb)	mg/kg	2	ND	ND	600#
镉 Cadmium(Cd)	mg/kg	2	ND	ND	75
汞 Mercury(Hg)	mg/kg	2	ND	ND	600
六价铬 Hexavalent Chromium(Cr(VI))	mg/kg	8	ND	ND	600##
一溴联苯 Mono-bromobiphenyl	mg/kg	5	ND	ND	—
二溴联苯 Di-bromobiphenyl	mg/kg	5	ND	ND	—
三溴联苯 Tri-bromobiphenyl	mg/kg	5	ND	ND	—
四溴联苯 Tetra-bromobiphenyl	mg/kg	5	ND	ND	—
五溴联苯 Penta-bromobiphenyl	mg/kg	5	ND	ND	—
六溴联苯 Hexa-bromobiphenyl	mg/kg	5	ND	ND	—
七溴联苯 Hepta-bromobiphenyl	mg/kg	5	ND	ND	—
八溴联苯 Octa-bromobiphenyl	mg/kg	5	ND	ND	—
九溴联苯 Nona-bromobiphenyl	mg/kg	5	ND	ND	—
十溴联苯 Deca-bromobiphenyl	mg/kg	5	ND	ND	—
多溴联苯 Polybrominated Biphenyls(PBBs)	mg/kg	—	ND	ND	600
一溴二苯醚 Mono-bromodiphenyl ether	mg/kg	5	ND	ND	—
二溴二苯醚 Di-bromodiphenyl ether	mg/kg	5	ND	ND	—
三溴二苯醚 Tri-bromodiphenyl ether	mg/kg	5	ND	ND	—
四溴二苯醚 Tetra-bromodiphenyl ether	mg/kg	5	ND	ND	—
五溴二苯醚 Penta-bromodiphenyl ether	mg/kg	5	ND	ND	—
六溴二苯醚 Hexa-bromodiphenyl ether	mg/kg	5	ND	ND	—
七溴二苯醚 Hepta-bromodiphenyl ether	mg/kg	5	ND	ND	—
八溴二苯醚 Octa-bromodiphenyl ether	mg/kg	5	ND	ND	—
九溴二苯醚 Nona-bromodiphenyl ether	mg/kg	5	ND	ND	—
十溴二苯醚 Deca-bromodiphenyl ether	mg/kg	5	ND	ND	—
多溴二苯醚 Polybrominated Diphenyl Ethers(PBDEs)	mg/kg	—	ND	ND	600
邻苯二甲酸二正丁酯 Dibutyl phthalate(DBP)	mg/kg	30	ND	ND	600
邻苯二甲酸丁苄酯 Butylbenzyl phthalate(BBP)	mg/kg	30	ND	ND	600
邻苯二甲酸二(2-乙基己基)酯 Di-(2-ethylhexyl) phthalate(DEHP)	mg/kg	30	ND	ND	600
邻苯二甲酸二异丁酯 Di-iso-butyl phthalate(DIBP)	mg/kg	30	ND	ND	600

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欧盟 RoHS-铅、镉、汞、六价铬 的含量/ EU RoHS-Pb, Cd, Hg, Cr(VI) content

检测方法/仪器/ Test Method/Equipment: IEC 62321-5:2013, IEC 62321-4:2013+AMD1:2017, IEC 62321-7-1:2015; AAS, ICP-OES, UV-VIS

检测项目 Test Item(s)	单位 Unit	方法检出限 MDL	结果 Result(s)		限值 Limit
			1-1	1-4	
铅 Lead(Pb)	mg/kg	2	14	ND	600#
镉 Cadmium(Cd)	mg/kg	2	ND	ND	75
汞 Mercury(Hg)	mg/kg	2	ND	ND	600
六价铬 Hexavalent Chromium(Cr(VI))	µg/cm ²	0.1	ND	ND	◆

检测项目 Test Item(s)	单位 Unit	方法检出限 MDL	结果 Result(s)		限值 Limit
			1-5	1-6	
铅 Lead(Pb)	mg/kg	2	71	ND	600#
镉 Cadmium(Cd)	mg/kg	2	9	ND	75
汞 Mercury(Hg)	mg/kg	2	ND	ND	600
六价铬 Hexavalent Chromium(Cr(VI))	µg/cm ²	0.1	ND	ND	◆

备注/ Note:

对于检测铅, 镉, 汞, 样品已完全消解。/ The sample(s) had been dissolved totally tested for Lead, Cadmium, Mercury.

µg/cm²=微克每平方厘米/ µg/cm²=microgram per square centimeter

mg/kg (毫克每千克) = ppm (百万分之)/ mg/kg (milligram per kilogram) = ppm (parts per million)

ND=未检出/ ND=Not Detected

MDL=Method Detection Limit

结果显示为 ND 不计入总和的计算。/ Results shown as ND are ignored in the sum calculation.

检测的铬(Cr)含量是“ND”, 则六价铬(Cr(VI))含量也是“ND”, 不需要进行六价铬(Cr(VI))的确认性测试。若铬(Cr)含量超过六价铬(Cr(VI))方法检出限, 需要进行六价铬(Cr(VI))的确认性测试。/ The detected Chromium (Cr) content is "ND", therefore, the Hexavalent Chromium (Cr (VI)) content is "ND", No need for validation test of the Hexavalent Chromium (Cr (VI)). If Chromium (Cr) content exceeds Hexavalent Chromium (Cr (VI)) method detection limit, Validation test of the Hexavalent Chromium (Cr (VI)) is required.

##=三价铬表面处理和热镀锌板镀层≤50mg/kg, 电镀彩锌表面和电镀白锌表面以及其他无铬处理镀层

ND, 其他物料≤600mg/kg。/ ##=Trivalent chromium surface treatment and hot dip galvanized sheet plating ≤50mg/kg, electroplating color zinc surface and electroplating white zinc surface and other chromium-free treatment plating ND, other materials ≤600mg/kg.

#=PVC 类电源线≤300mg/kg, 钢合金≤3200mg/kg, 铝合金≤3600mg/kg, 铜合金≤36000mg/kg, 其他物料≤600mg/kg。/ #=PVC category power cord≤300mg/kg, steel alloy≤3200mg/kg, aluminum alloy≤3600mg/kg, copper alloy≤36000mg/kg, other material≤600mg/kg.

◆ = 按照检测方法 IEC 62321-7-1:2015: ①当六价铬的浓度高于 0.13µg/cm²时, 样品为阳性, 即含有六价铬; ②当六价铬的浓度为 ND(低于 0.10µg/cm²)时, 样品为阴性, 即未检测到六价铬; ③当六价铬的浓度介于 0.10µg/cm²与



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0.13 $\mu\text{g}/\text{cm}^2$ 之间时，无法直接判定是否检测到六价铬；因不同个体的样品表面差异可能会影响测定结果；由于未获知样品的存储条件和生产日期，样品的六价铬检测结果仅能代表检测时样品含六价铬的状态。 / ◆ = According to the test method IEC 62321-7-1:2015:①The sample is positive for Cr(VI) if the Cr(VI) concentration is greater than 0.13 $\mu\text{g}/\text{cm}^2$. The sample coating is considered to contain Cr(VI); ②The sample is negative for Cr(VI) if Cr(VI) is ND (concentration less than 0.10 $\mu\text{g}/\text{cm}^2$). The coating is considered a non-Cr(VI) based coating; ③The result between 0.10 $\mu\text{g}/\text{cm}^2$ and 0.13 $\mu\text{g}/\text{cm}^2$ is considered to be inconclusive -unavoidable coating variations may influence the determination; The surface difference of samples from different individuals may affect the determination results of Cr(VI); Information on storage conditions and production date of the tested sample is unavailable and thus Cr(VI) results represent status of the sample at the time of testing.inconclusive -unavoidable coating variations may influence the determination; Information on storage conditions and production date of the tested sample is unavailable and thus Cr(VI) results represent status of the sample at the time of testing.

样品描述 / Sample Description:

序号 No.	HCT 样品 ID HCT Sample ID	检测点描述 Test Part Description			备注 Note
1-1	WTF24H04091192C.1	1-1	红印银色金属外壳	Red printed silver metal shell	●
1-2		1-2	黑色软胶塞	Black soft plastic stopper	●
1-3		1-3	黑色湿纸	Black electrolytic paper	●
1-4		1-4	暗银色金属箔	Dull silver metal foil	●
1-5		1-5	银色金属箔	Silver metal foil	●
1-6		1-6	银色金属引脚	Silver metal pin	●

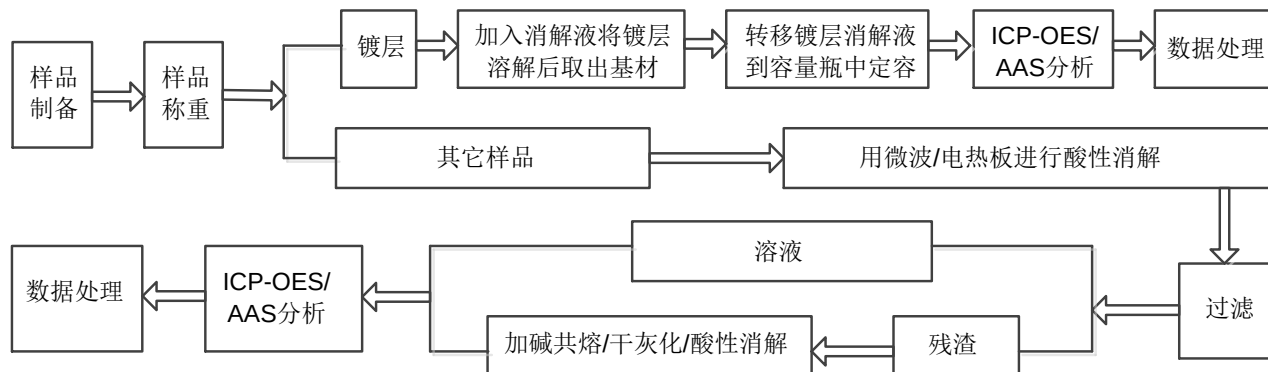
备注 / Note:

●=实际检测样品/●=Actual tested sample

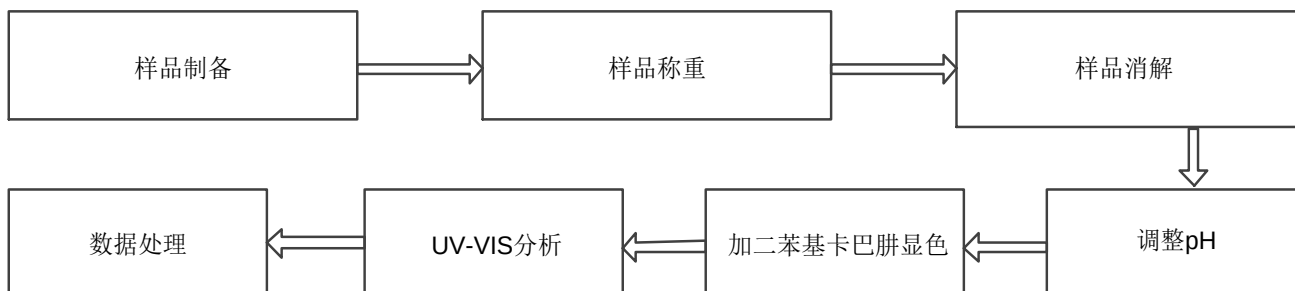
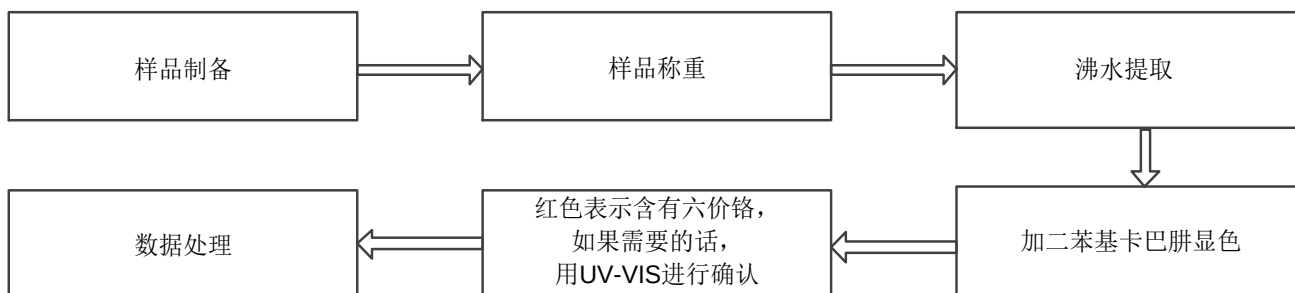


检测流程图 / Test Flow Chart:

铅(Pb)、镉(Cd)、汞(Hg)、铬(Cr)



六价铬(Cr(VI))

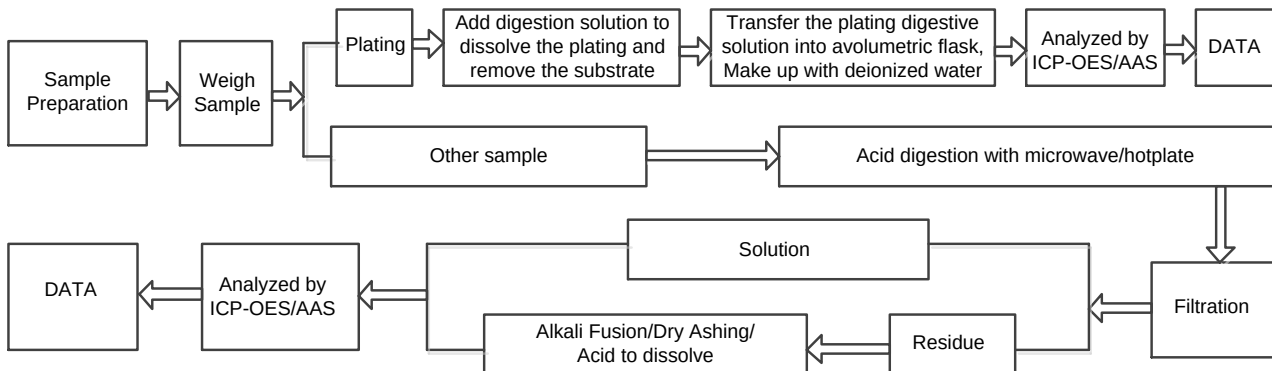


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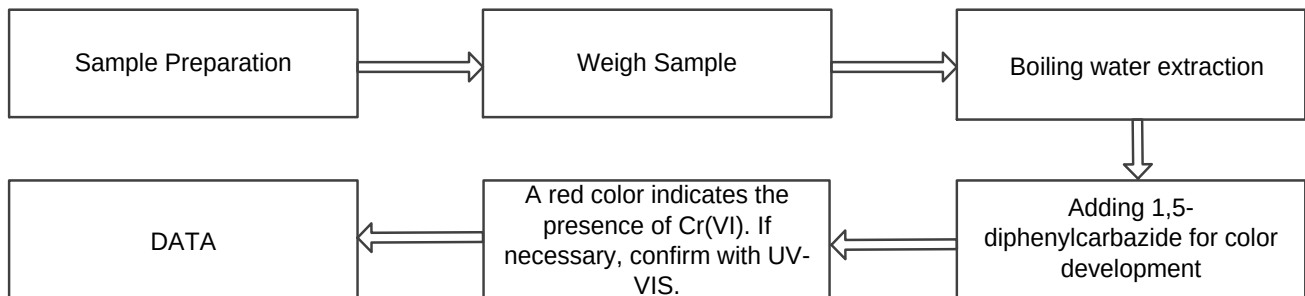
多溴联苯(PBBs)、多溴二苯醚(PBDEs)、邻苯二甲酸二正丁酯(DBP)、邻苯二甲酸丁苄酯(BBP)、邻苯二甲酸二(2-乙基己基)酯(DEHP)、邻苯二甲酸二异丁酯(DIBP)



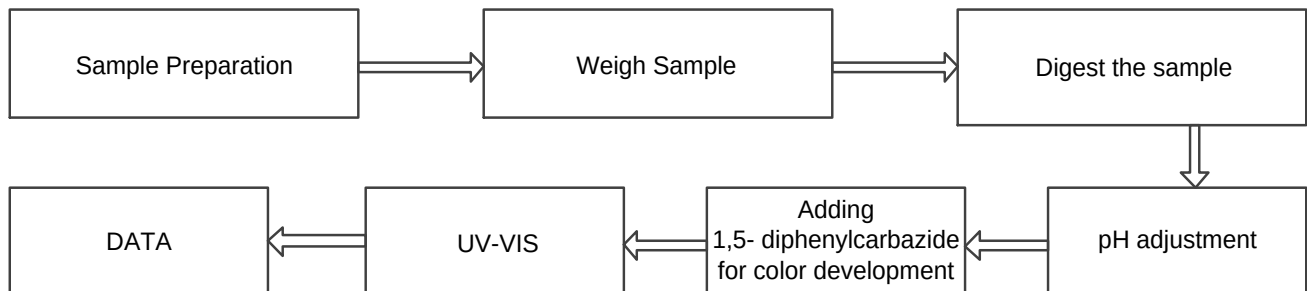
Lead(Pb), Cadmium(Cd), Mercury(Hg), Chromium(Cr)



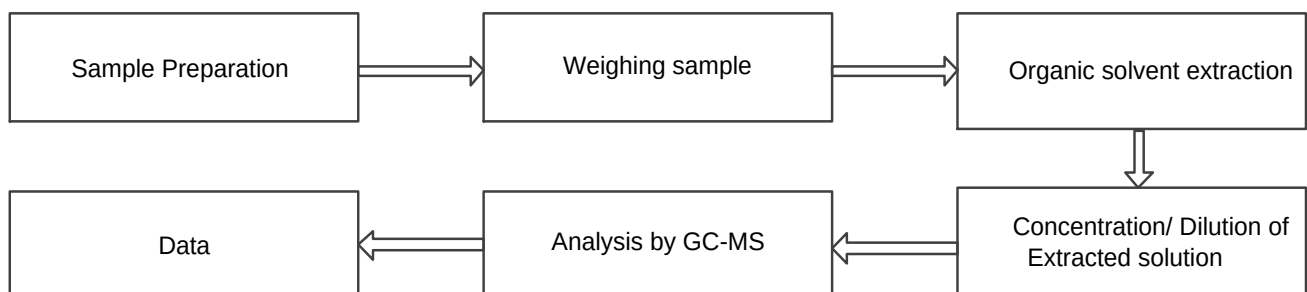
Hexavalent Chromium(Cr(VI))



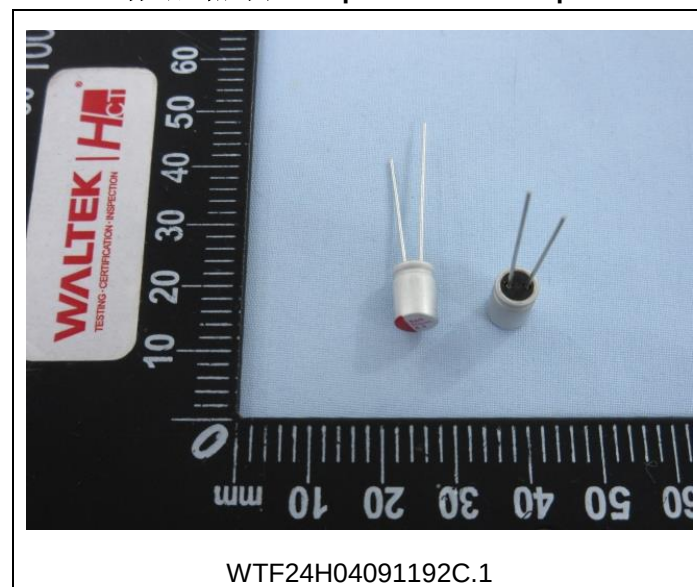
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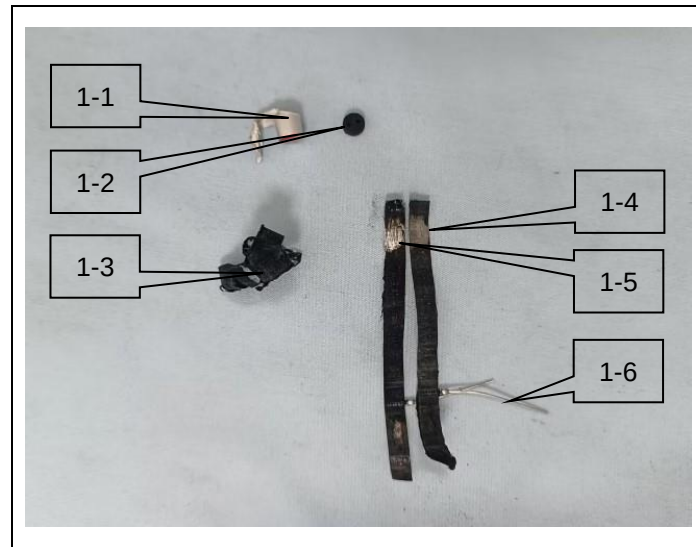
Polybrominated Biphenyls(PBBs), Polybrominated Diphenyl Ethers(PBDEs), Dibutyl phthalate(DBP), Butylbenzyl phthalate(BBP), Di-(2-ethylhexyl) phthalate(DEHP), Di-iso-butyl phthalate(DIBP)



样品附图/ The photo of the sample



拆分附图/ The photo of Disassembly



声明/ Statement:

1. 检测报告无批准人签字和专用章无效。
This report is considered invalid without approved signature and special seal.
2. 委托单位及地址，样品和样品信息由委托方提供，委托方应对其真实性负责，HCT 未核实其真实性。
The Applicant name and Address, the sample(s) and sample information was/were provided by the applicant who should be responsible for the authenticity which HCT hasn't verified.
3. 本报告检测结果（结论）仅对受测样品负责。
The result(s)(conclusion) shown in this report refer(s) only to the sample(s) tested.
4. 未经 HCT 书面同意，不得部分复制本报告。
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The result(s) in no CMA logo report shall only be used for client's scientific research, teaching, internal quality control, product research and development, etc..and just for internal reference.
6. 有 CNAS 标识报告中的“n”代表该检测项目暂未申请 CNAS 认可。
The “n” in CNAS logo report means that the test item(s) was (were) currently not applying for CNAS accreditation.
7. 本报告使用的判定规则：
Decision rules used in this report:
(1)按照检测要求列的法规/标准中规定的判定规则；
(2)如果检测要求列的法规中没有规定判定规则的话，则按照《CNAS-GL015 判定规则和符合性声明指南》6.2.1 简单接受(w=0)的判定规则：
合格(接受)--测得值位于容许区间以内。
不合格(拒绝)--测得值位于容许区间以外。
(1)According to the Decision rules in the regulations/standards listed in the Test Requested;

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(2) If there is no Decision rules specified in the regulations listed in the Test Requested, then according to CNAS-GL015 Guidelines on Decision Rules and Statements of Conformity, 6.2.1, Simple Acceptance ($w=0$) of The binary Decision rule:

PASS (Accepted) - The measured value is within the tolerance interval.

FAIL (Rejected) - The measured value is outside the tolerance interval.

=====报告结束/ End of Report =====

